

Hou

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EDUCATION

University of Illinois Urbana-Champaign
Bachelor of Science in Computer Engineering

Champaign, IL
Expected May 2026

Cumulative GPA: 3.85/4.00

Relevant Courses: Machine Learning, Database System, Computer Systems & Programming, Linear Algebra with Computational Applications, Probability with Engineering Applications, Introduction to Computing

RESEARCH

Contrastive Label Disambiguation for Partial Label Learning

Supervised by Prof. Lu (Huazhong University of Science and Technology)

May 2024 - Present

- Summarized existing research methods for partial label learning, with a focus on models like PiCo, PiCo+ using Contrastive Learning
- Proposed WUCO, a weakly supervised contrastive learning framework designed to identify potential false-negative samples in X-ray images that were initially classified as negative
- Implemented sliding label disambiguation by measuring the distance between the negative sample features generated through weakly supervised contrastive learning and the class feature prototype of the positive category

Image Quality Improvement by Super Resolution Generative Adversarial Networks (SRGAN)

Supervised by Prof. Mark Vogelsberger (Massachusetts Institute of Technology)

Jun 2021 - Sep 2021

- Reviewed academic literature on machine learning algorithms, including GAN and SRGAN network
- Collaborated with a team of 4 to develop, train (using 800+ images), and test an SRGAN model to improve digital image resolution
- Co-authored a 12-page research paper published in IEEE Xplore (DOI: 10.1109/ICHCI54629.2021.00031)

PROJECT

Implementation of Mathematical Theories

Feb 2023 - May 2023

- Worked on the mathematical theory of linear algebra and implemented it through **Python** programming labs
- Completed various linear algebra-related projects, including implementing Google's PageRank algorithm, developing image and audio compression schemes like JPEG and MP3
- Realized automatic face recognition alongside other machine learning algorithms such as linear regression

Sound-controlled Robot

Feb 2023 - May 2023

- Led a team of 4 to design and build a sound-controlled robot car, managed documentation process for the project, and streamlined teamwork efficiency using Notability
- Built the robot's circuit and basic structure, including plug-and-play components and wheels
- Provided feedback on the design style and participated in the editing process for final presentation video
- Earned full grade plus extra credit in the course Introduction to Electronics

Logistics for a Vending Machine Project

Sep 2022 - Dec 2022

- Designed and developed a vending machine hardware and software system using C/C++ and assembly language to achieve LC3 specifications
- Implemented calculations and deductions for vending machine transactions
- Successfully detected coins, calculated the amount of money added, and provided feedback to the user

Text-based Game Development

Nov 2021 - Dec 2022

- Used **Java** to create a text-based game that automates the process of creating narratives, allowing users to enter a basic set of text to generate a personalized story

SUMMARY

Innovative and goal-oriented student with strong passion for machine learning and experience in technical problem-solving and algorithmic research, skilled in team building with growth mindset. **Actively seeking internship opportunities to work as a software engineer.**

- Programming Languages: Python, Java, C, C++, LC3, Office Suite
- Machine Learning Models: SRGAN, GAN, CNN